

4x4 Accelerator Controller

v1.0 Specification - Systolic Array Control Path

1. Purpose

This document defines the v1.0 control-path specification for the 4x4 signed INT8 systolic-array accelerator. The controller sequences accumulator clearing, four matrix feed cycles, wavefront draining, and completion signaling. It does not perform arithmetic; matrix multiplication remains in the datapath.

2. Module Interface

Signal	Dir.	Width	Description
clk	Input	1	Clock
rst_n	Input	1	Asynchronous active-low reset
start	Input	1	Starts one operation when accepted in IDLE
busy	Output	1	High during CLEAR, FEED, and DRAIN
done	Output	1	One-cycle completion pulse in DONE
clear_acc	Output	1	High only in CLEAR; clears all PE accumulators
feed_valid	Output	1	High only in FEED; enables A/B data injection
k_counter	Output	2	Selects k = 0,1,2,3 during FEED

3. FSM

The controller contains five states:

IDLE -> CLEAR -> FEED -> DRAIN -> DONE -> IDLE

State	Duration	busy	clear_acc	feed_valid	done	Function
IDLE	Until start	0	0	0	0	Wait for a new operation
CLEAR	1 cycle	1	1	0	0	Clear all 16 PE accumulators
FEED	4 cycles	1	0	1	0	Stream k=0,1,2,3 operands
DRAIN	6 cycles	1	0	0	0	Allow final wavefront to reach PE(3,3)
DONE	1 cycle	0	0	0	1	Completion pulse

4. Start / Busy / Done Protocol

start is accepted only while the controller is in IDLE. A start presented while busy is ignored in v1.0. **busy** is asserted throughout CLEAR, FEED, and DRAIN. **done** is asserted for exactly one cycle in DONE. After DONE, the controller returns to IDLE and can accept the next operation.

5. FEED Operation and k_counter

FEED lasts exactly four cycles. k_counter identifies which inner-product term is injected into the datapath. The matrix-storage/read logic uses the counter as follows:

```
a_feed[i] = A[i][k_counter]
b_feed[j] = B[k_counter][j]
```

FEED cycle	k_counter	A data	B data
1	0	A[:,0]	B[0,:]
2	1	A[:,1]	B[1,:]
3	2	A[:,2]	B[2,:]
4	3	A[:,3]	B[3,:]

6. DRAIN Latency

For PE(i,j), inner-product term k is consumed at wavefront edge E(k+i+j), taking E0 as the edge that consumes k=0 at PE(0,0). The latest operation is k=3 at PE(3,3): $3 + 3 + 3 = 9$. The final feed k=3 occurs at E3, so six additional rising-edge intervals are required before the final MAC completes at E9.

```
DRAIN_CYCLES = 6
```

Edge	Controller activity	Meaning
E0	FEED k=0	First inner-product term enters
E1	FEED k=1	
E2	FEED k=2	
E3	FEED k=3	Last external feed
E4	DRAIN 1	
E5	DRAIN 2	
E6	DRAIN 3	
E7	DRAIN 4	
E8	DRAIN 5	
E9	DRAIN 6	PE(3,3) completes final MAC
E10	DONE	Result is complete; done pulse

7. Counter Requirements

k_counter: 2 bits; meaningful during FEED and sequences 0 -> 1 -> 2 -> 3. **drain_counter**: internal counter sufficient to represent six DRAIN cycles. Counter initialization and terminal-count logic must avoid an off-by-one transition that would assert DONE before the E9 accumulation is committed.

8. Reset Behavior

When rst_n is asserted low, the controller returns to IDLE and control outputs are inactive: busy=0, done=0, clear_acc=0, feed_valid=0. Counters return to defined reset values. Reset behavior must be consistent with the

asynchronous active-low reset convention already used by the PE and input-skew RTL.

9. v1.0 Datapath Integration

For v1.0, A and B may be stored in internal 4x4 register arrays rather than SRAM. The controller supplies k_counter; combinational read/select logic forms A[:,k] and B[k,:], which then enter input_skew and the 4x4 systolic PE array.

Matrix registers -> k_counter address/select -> input_skew -> 4x4 PE array -> C[4][4]

10. v1.0 Scope / Non-Goals

Included: one-operation start protocol, busy/done status, accumulator clear, four-cycle feed sequencing, exact six-cycle drain, and deterministic control. Not included in v1.0: SRAM interface, DMA, ready/valid command queueing, backpressure, multiple outstanding matrices, tiling, variable matrix size, or UVM.

11. Acceptance Criteria

Requirement	Expected behavior
Reset	Controller returns to IDLE with inactive control outputs
Start	Accepted only in IDLE
CLEAR	Exactly 1 cycle
FEED	Exactly 4 cycles; k_counter = 0,1,2,3
DRAIN	Exactly 6 cycles after the final feed edge
DONE	Exactly 1 cycle
busy	High only during CLEAR, FEED, DRAIN
Repeated start while busy	Ignored in v1.0
Datapath result	C is complete before/when DONE is presented

Version: v1.0 | Project: 4x4_accelerator | Arithmetic datapath: signed INT8 inputs, 32-bit PE accumulators